PROJECT REPORT

JanaVignanam

MediaSphere constituency intelligence platform

A complete technical and functional report of the Narasaraopet Lok Sabha workspace: news pipeline, website, admin desk, WhatsApp bot, grievances, visits, and birthday wishes.

Field	Detail
Product name	JanaVignanam (workspace) / MediaSphere (engineering name)
Constituency	Narasaraopet Lok Sabha, Palnadu district, Andhra Pradesh
Member of Parliament	Sri Lavu Sri Krishna Devarayulu
Report date	24 October 2026
Code base	GitHub NaiduBugata/MediaSphere and NaiduBugata/janavignanam
Website	Vite React app on Vercel (client/)
API	NestJS on Railway (server/), health GET /api/health
Database	Neon PostgreSQL schema mediasphere.documents (JSON documents)
Messaging	Meta WhatsApp Cloud API

1. Purpose

JanaVignanam is the constituency operating system for Narasaraopet. It collects local news, stores grievances and visits, and lets the MP and staff read that same information on the website and on WhatsApp. The public-facing product name on the login screen is JanaVignanam. The repository and API still use the name MediaSphere.

The system is built so a staff member does not need to open five different tools. News arrives on an hourly cycle. A person can file a grievance by sending WhatsApp text. An MP or super admin can open a menu on WhatsApp and ask for news, visits, grievances, projects, constituency facts, campaigns, analytics, or follow-up contacts. The admin desk at /@admin is the control room for fetch, uploads, visits, birthdays, outbound WhatsApp, MP access, and the demo-data switch.

2. What the product is for

- Keep Narasaraopet news in one place, tagged by place, sentiment, and whether action is needed.
- Let citizens (role person) send a WhatsApp message that becomes an Open grievance on the website.
- Let the MP and super admin browse live sections from WhatsApp without leaving the chat.
- Remind visit leads up to three times before an event, and wish saved contacts on their birthday.
- Give the office a workspace for grievances, projects, constituency contacts, and campaigns.
- Give administrators a strict upload path so only files that match the stored format are accepted.

3. System architecture

The product is two deployable services from one Git repository. They never share a process. The website is a static Vite build. It talks to the API over HTTPS. The API owns the database, the news cycle, email reports, and every WhatsApp send and receive.

Piece	Technology	Host	Role
Website	React 19, Vite 8, Tailwind 4, React Router 7	Vercel (client/)	Workspace login, news desk, admin desk
API	NestJS 11, Node, TypeScript	Railway (server/, Dockerfile)	All business logic and schedulers
Database	Neon Postgres, table mediasphere.documents	Neon	Articles, contacts, visits, grievances, settings
AI	Groq	External	Article summaries and WhatsApp news wording
News sources	Lokal, Sakshi, YouTube	External	Hourly constituency collection
WhatsApp	Meta Cloud API (Graph)	Meta	Inbound webhook and outbound lists/text
Email	Resend (optional)	External	Digest, failure, daily report

Figure-equivalent: browser → Vercel (client) → Railway /api → Neon. Meta POSTs to Railway /webhook (no /api prefix). Railway health check is GET /api/health.

3.1 Data store

MongoDB is no longer on the read/write path. DatabaseService is a PostgreSQL pool with ssl rejectUnauthorized true. Collections are JSON rows in mediasphere.documents (collection, doc_id, doc jsonb). A thin PgDocumentCollection wrapper keeps a Mongo-shaped API (find, findOne, insertOne, updateOne) so existing

services did not need a rewrite. The health JSON still exposes a field named mongo; true means Neon answered.

Collection	What it holds
articles	News items from the pipeline (Lokal, Sakshi, YouTube)
jv_records	Workspace rows: visits, grievances, campaigns, projects, people
jv_users	Workspace login accounts (email, scrypt password)
birthday_contacts	Staff, MP, persons, users: phone, role, birthday, allowReplies
whatsapp_webhook_events	Inbound messages and delivery statuses
app_settings	Flags such as demo-data enabled
visit_files	Separate table mediasphere.visit_files for PDF/Word/Excel bytes

4. Website (client)

The Vite app has two shells. Most routes go through App.tsx, which requires a workspace login stored in localStorage key janavignanam.session. The news desk is embedded when the user opens News. A separate route /@admin is the operations console and uses admin credentials, not the workspace account.

4.1 Workspace (after login)

Route	Screen	Data
/	Command center	Counts of open items, saved records, news articles
/grievances	Grievances	jv_records section grievances; Reviewed plus follow-up department
/projects	Projects & reports	Delivery rows; demo catalog if the switch is on and none stored
/news	News desk	Articles with filters, featured story, visits strip
/people	Constituency	Assembly segments and coordinators
/campaigns	Campaigns	Plans only; this screen does not send WhatsApp
/analytics	Analytics	Article totals, sentiment, problem counts

Both the workspace login and the admin login show a password visibility toggle and a signing-in / creating-account state after submit.

4.2 News desk

Inside /news the layout is editorial: trending bar, featured story, top stories rail, latest row, then a card grid or table. Filters cover search, source, category, sentiment, priority, district, mandal, village, and date range. A visits section lists published visits from GET /api/visits (lead phones are omitted on the public list).

4.3 Admin desk (/@admin)

Tab	What it does
Fetch monitoring	Pipeline status, history, and a manual fetch when authorised
Upload	Strict file upload for visits, schedule, campaigns, grievances, projects, constituency, leaders, birthdays. HTTP success only if every row is valid; otherwise nothing is inserted

Tab	What it does
Visits	Bulk import or a single visit; lead WhatsApp is stored for reminders and hidden from the site
Birthdays	Contact list and a manual wish send
Send WhatsApp	Session text or an approved template to a chosen number
MP access	Grant Sri Lavu Sri Krishna Devarayulu access to a new number so the bot answers at once; demo-data switch

5. API (server)

NestJS boots in server/src/main.ts. Helmet is on, CORS is limited to configured origins plus optional *.vercel.app previews, and class-validator is global. Default local port is 5000; Railway injects PORT. Do not set PIPELINE_ON_API=true on a laptop while production is already running the hourly cycle.

5.1 Nest modules

Module	Responsibility
HealthModule	GET /api/health and /api/database/health
NewsModule	GET /api/news and /api/news/stats
PipelineModule	Hourly native fetch: Lokal, Sakshi, YouTube; lock; admin trigger
ReportsModule	Daily report email at 07:00 Asia/Kolkata
WhatsAppModule	Webhook GET/POST /webhook; outbound Graph sends
Chatbot (inside WhatsApp)	Menu, grievances intake, follow-up contacts, news answers
WorkspaceModule	Register/login, records, summary
AdminModule	Auth, fetch console, upload, MP access, demo switch
VisitsModule	Public and admin visit lists, files, import, reminders
BirthdaysModule	Contacts, muted senders, 15-minute wish cron
NotificationsModule	Email and WhatsApp channel status for the UI

5.2 News pipeline

When PIPELINE_ON_API=true and PIPELINE_EXECUTOR=native, the API runs the collection cycle on an interval (default one hour) with a database lock (TTL about 45 minutes). PIPELINE_CATCHUP_ON_START should be false in production so a Railway restart does not fire an extra fetch. Sources: Lokal local news, Sakshi headlines, YouTube captions for constituency channels. Articles are kept only when they match Narasaraopet assembly geography. Groq writes a complete 45–65 word summary that must not end mid-sentence.

5.3 Reports and email

If REPORT_ENABLED and the report scheduler are on, a daily pack is emailed at 07:00 IST. Automatic WhatsApp alerts (news digest, pipeline failure, daily summary) stay behind WHATSAPP_ALERTS_ENABLED and by default go to email instead. Session WhatsApp replies from the bot are separate and do not use that flag.

6. WhatsApp

6.1 Flow

Meta sends POST /webhook to Railway (no /api prefix). The controller records inbound time on known contacts, drops muted (one-way birthday) senders, stores the event in Neon, then asks ChatbotService.consider. Outbound calls go to graph.facebook.com/{version}/{WHATSAPP_PHONE_NUMBER_ID}/messages. The website itself never talks to Meta except through the admin Send WhatsApp tab, which calls POST /api/admin/messages.

6.2 Who is answered

Role	Typical people	Bot behaviour
superadmin	Sarojininaidu	Full menu. Pipeline success/failure WhatsApp. Never muted.
admin	Akshay Guptha	Normal menu (top 5 items). Alerts with staff. Never muted.
mp	MP numbers from MP access; also existing MP contacts	Full menu (top 6). No automatic pipeline alerts. Never muted.
person	Udatha Sravani, Balaji	No menu. Their text is saved as a grievance and they get a receipt.
user	Test rows, one-way contacts	Muted if allowReplies is false; message dropped before the chatbot.

Staff directory is loaded from birthday_contacts. Granting MP access writes role mp, allowReplies true, mpAccess true under the name Sri Lavu Sri Krishna Devarayulu, then reloads the in-process directory so the next webhook is answered without a restart. A number that already belongs to admin, superadmin, person, or another MP is refused.

6.3 Menu

The control is a WhatsApp interactive list. The button label is Menu. There is no header and no row description on the main menu, so the echo stays a single line. WhatsApp still requires select then Send; that is a Meta rule. Sections: News, Visits, Followup, Grievances, Projects & reports, Constituency, Campaigns, Analytics. Visit screens prepend Follow up (visit reminders, not grievance follow-up).

List body is capped at 1024 characters. News, visits, projects, constituency, campaigns, analytics, and grievances stay in that one list with Menu. If the text is longer, whole items that still fit are kept; leftover “Read more” is stripped. WhatsApp allows at most ten tappable rows, so a Followup directory of 25 names is sent as three consecutive lists.

6.4 Grievances on WhatsApp

A person-role number is never given the menu. Their message is stored as title (first 80 characters), full detail, place, department, leader, language, priority, status Open, createdBy, reporterPhone, sourceMessageId. English script gets an English receipt; Telugu script gets Telugu. Romanized Telugu (for example “maa gram lo neellu levu”) is interpreted only for priority and department; the original words are stored and shown to the MP.

Staff opening Grievances see open items (reviewed rows are hidden). Choosing one shows the issue plus Follow up and the main sections. Follow up sends: (1) a department order to the contact whose name contains “akshay”; (2) a leader notice to that leader’s phone or Akshay if the mock leader has no number; (3) a progress note to the filer in the issue language. The row is marked Reviewed and leaves the WhatsApp list. The website shows “Reviewed. Follow up sent to {department}.” Mock departments (water, roads, electricity, health, revenue, else

general administration) are names only.

6.5 Followup section (contacts)

A dedicated Followup item opens three lists: Recent Follow-ups (25 Indian names), Frequently Followed Up (25 names), Not followed up (8 names). Every mock contact uses 891 953 7879. Tapping a name sends a follow-up text to that number. The first send moves the person to Recent; a later send moves them to Frequently Followed Up.

6.6 Visits and birthdays

Automatic visit reminders go only to that visit's lead, at most three times (timed visits: 3/2/1 hours before; anytime visits: 07:00, 12:00, 16:00 IST). Menu Follow up on Visits messages every lead on every saved visit. Birthday wishes run every 15 minutes in Asia/Kolkata from wish hour (default 7) until 21:00. A contact is due when MM-DD matches today and lastWish is not already for that ISO day. Local .env often has BIRTHDAY_WISHES_ENABLED=false so a laptop API does not send live wishes.

7. Demo data

The admin MP access tab has a Demo data switch. On: mock visits (title MOCK… or detail MOCK-VISIT), mock grievances, campaigns, constituency people, and in-memory project samples are visible on WhatsApp and on the website. Off: those rows are hidden but not deleted. Visit reminder jobs are not stopped. The flag is stored in app_settings key demo-data and loaded when the API starts.

8. Security and operations

- Never commit server/.env or client/.env. Tokens, Neon URLs, Meta secrets, and admin passwords stay in host variable screens.
- Admin routes use AdminOnly / bearer or X-Admin-Token. Workspace routes use a 12-hour HMAC session.
- CORS allow-list is the Vercel origin. Optional preview hosts matching *.vercel.app can be enabled.
- Neon SSL is verify-full (rejectUnauthorized true). Do not weaken it.
- WhatsApp phone numbers in logs are masked (first two and last two digits).
- Muted senders are dropped before storage so one-way birthday contacts cannot drive the bot.
- Railway builds the Docker image from server/. GET /api/health must return 200.
- Do not start a local API with PIPELINE_ON_API=true while production is already fetching.

9. Deploy

Step	Action
1	Push main to GitHub (janavignanam for Railway; MediaSphere as the origin copy).
2	Railway: root directory server, Dockerfile, health GET /api/health.
3	Set NEON_DATABASE_URL, CORS_ORIGINS, admin secrets, pipeline flags, WhatsApp, Groq.
4	Vercel: root client/, VITE_API_BASE_URL = Railway origin, then redeploy.
5	Meta webhook URL = https://<railway-host>/webhook.

A commit that exists only on the laptop will not appear on Railway. Both remotes must receive the same SHA. Force-push is not used for normal releases.

10. Known limits

- WhatsApp list messages: maximum 10 rows and 1024 characters of body. Twenty-five contacts cannot be one tappable list.
- Meta still requires Send after choosing a list row.
- Template sends can fail with Meta 131042 (business eligibility / payment). Session text inside 24 hours still works.
- Neon free-plan CU-hours can suspend compute. The website then shows database not connected and WhatsApp says Nothing saved yet even though rows are still stored.
- Health field mongo means Neon connectivity, not MongoDB.
- Mock department leaders have names only; leader notices fall back to Akshay’s number.
- Bobby is not in birthday_contacts unless added; mock grievance notes to Bobby cannot deliver until a contact name contains bobby.
- In-memory flags (demo switch, follow-up contact moves, staff directory) apply to the running Railway process. A second instance would need its own load.

11. Empty screens and “database not connected”

The workspace header reads Database connected only when GET /api/workspace/summary returns mongo: true. That probe is Neon. If Neon has exceeded quota, the SSL handshake fails, or NEON_DATABASE_URL is missing on Railway, the API still serves /api/health with mongo: false. The website then shows empty lists. WhatsApp formatSection prints Nothing saved yet when sectionRecords cannot read jv_records.

When Demo data is on, the chatbot fills Projects, Visits, Campaigns, Constituency, Grievances, and Followup from built-in sample catalogs if the database returns nothing. That keeps WhatsApp usable during a Neon outage. The website still needs a healthy API and a healthy Neon connection to list real articles and saved records. Turning Demo data off hides MOCK rows on purpose.

Fix path: confirm Railway GET /api/health, then GET /api/database/health, then Neon project status (compute not suspended). Confirm Vercel VITE_API_BASE_URL points at that Railway origin and CORS_ORIGINS includes the Vercel host. Do not point the website at a laptop API.

12. Repository map

Path	Contents
client/src/App.tsx	Workspace login and section router
client/src/mediasphere/	News desk, admin page, visits, birthdays, upload, MP access
client/src/lib/api.ts	Workspace HTTP helper
server/src/main.ts	HTTP bootstrap, CORS, validation
server/src/database/	Neon pool and JSON document collections
server/src/pipeline/	Hourly news cycle
server/src/chatbot/	WhatsApp menu, grievances, follow-up contacts
server/src/whatsapp/	Webhook, Graph send, audience, mute filter
server/src/admin/	Auth, fetch, upload, desk (MP + demo)

Path	Contents
server/src/visits/	Visit CRUD, import, reminder plan
server/src/birthdays/	Contacts and wish scheduler
server/railway.toml	Docker build and /api/health check
DEPLOYMENT.md	Railway and Vercel runbook

13. Recent delivery (selected)

Commit	Change
f18c5f4	Person intake, full MP menu, strict admin upload, password toggle
dd680e8	Analytics counts, Narasaraopet data, full news text, grievance language
463220f	Progress note to the filer; department order stays with Akshay
f609f4f	Reviewed grievances leave WhatsApp and show on the website
d350a97	MP access form, demo switch, leader notice on follow-up
0daaaf2	WhatsApp Followup directory and demo rows under the switch
7266944	News and Menu in one WhatsApp list (no second Choose a section)

14. Closing

JanaVignanam is a two-service product: a Vercel website and a Railway Nest API on Neon, with Meta WhatsApp as the field channel. The MP and super admin use the same sections on the phone and on the desk. Persons file grievances without seeing the menu. Demo data is a visibility switch, not a delete. Production behaviour depends on Neon remaining within quota and on GitHub main matching the SHA Railway actually built. This report describes the system as implemented in the Media repository on 24 October 2026. It does not print secrets, full phone numbers, or environment files.

End of report.